```
100 'CHEBYSHEV FILTER-  RECURSION COEFFICIENT CALCULATION
110 '
120                                         'INITIALIZE VARIABLES
130 DIM A[22]                               'holds the "a" coefficients upon program completion
140 DIM B[22]                               'holds the "b" coefficients upon program completion
150 DIM TA[22]                              'internal use for combining stages
160 DIM TB[22]                              'internal use for combining stages
170 '
180 FOR I% = 0 TO 22
190   A[I%] = 0
200   B[I%] = 0
210 NEXT I%
220 '
230 A[2] = 1
240 B[2] = 1
250 PI = 3.14159265
260                                         'ENTER THE FOUR FILTER PARAMETERS
270 INPUT "Enter cutoff frequency  (0 to .5):  ", FC
280 INPUT "Enter  0 for LP,  1 for HP filter:   ", LH
290 INPUT "Enter percent ripple    (0 to 29):   ", PR
300 INPUT "Enter number of poles (2,4,...20):              ", NP
310 '
320 FOR P% = 1 TO NP/2                      'LOOP FOR EACH POLE-PAIR
330   '
340   GOSUB 1000                            'The subroutine in TABLE 20-5
350   '
360   FOR I% = 0 TO 22                      'Add coefficients to the cascade
370     TA[I%] = A[I%]
380     TB[I%] = B[I%]
390   NEXT I%
400   '
410   FOR I% = 2 TO 22
420     A[I%] = A0*TA[I%] + A1*TA[I%-1] + A2*TA[I%-2]
430     B[I%] =      TB[I%] - B1*TB[I%-1] - B2*TB[I%-2]
440   NEXT I%
450   '
460 NEXT P%
470 '
480 B[2] = 0                                'Finish combining coefficients
490 FOR I% = 0 TO 20
500   A[I%] = A[I%+2]
510   B[I%] = -B[I%+2]
520 NEXT I%
530 '
540 SA = 0                                  'NORMALIZE THE GAIN
550 SB = 0
560 FOR I% = 0 TO 20
570   IF LH = 0 THEN SA   =  SA     +  A[I%]
580   IF LH = 0 THEN SB   =  SB     +  B[I%]
590   IF LH = 1 THEN SA   =  SA     +  A[I%] * (-1)^I%
600   IF LH = 1 THEN SB   =  SB     +  B[I%] * (-1)^I%
610 NEXT I%
620 '
630 GAIN = SA / (1 - SB)
640 '
650 FOR I% = 0 TO 20
660    A[I%] = A[I%] / GAIN
670 NEXT I%
680 '                                       'The final recursion coefficients are in A[ ] and B[ ]
690 END
```

TABLE 20-4